higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PRINCIPLES OF COMPUTER PROGRAMMING
(First Paper)
NQF LEVEL 3

22 NOVEMBER 2019

This marking guideline consists of 6 pages.

Copyright reserved

SECTION A

QUESTION 1

1.1 B
1.2 A
1.3 C
1.4 C
1.5 B
1.6 C
1.7 C
1.8 C
1.9 B
1.10 A

(10 × 1) **[10]**

QUESTION 2

2.1 False
2.2 True
2.3 True
2.4 True
2.5 False
2.6 False
2.7 False
2.8 False
2.9 True
2.10 False

(10 × 1) **[10]**

QUESTION 3

3.1 Algorithm
3.2 Octal
3.3 Database
3.4 Two-dimensional
3.5 Boolean

(5 × 1) **[5]**

TOTAL SECTION A: 25

SECTION B

QUESTION 4

4.1 Step 1: Change the zeros to ones and the ones to zeros.

10011001_2

01100110_2✓

Step 2: Add 1 to the answer above.

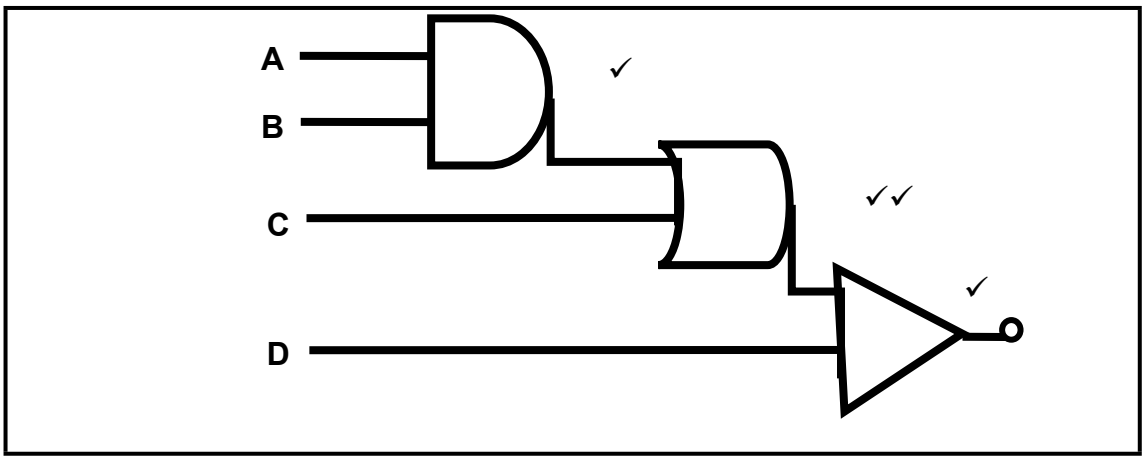
01100110_2

\+ 1_2✓

Step 3: Write the answer down.

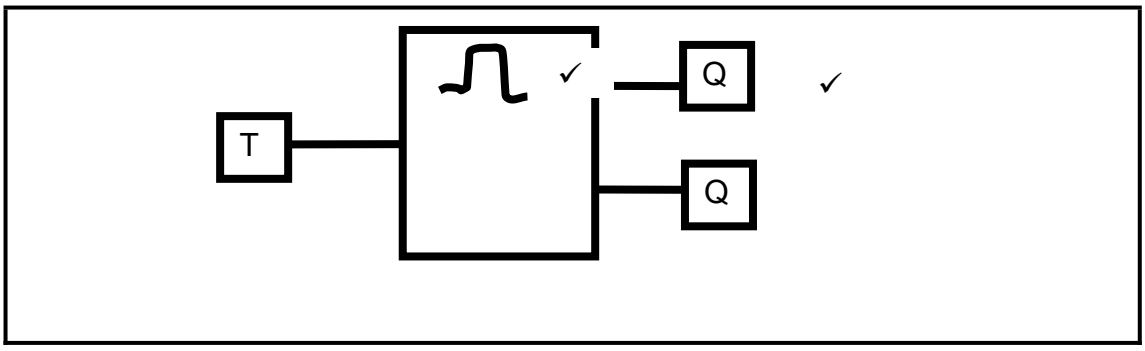
01100111_2✓✓ (4)

4.2



(4)

4.3



(2)
[10]

QUESTION 5

5.1 Do substitution

A < 5 and B = 10 or NOT D and C > A or NOT E

-2 < 5 and 10 = 10 or NOT TRUE and 3 > -2 or NOT FALSE✓✓

Simplify

TRUE and TRUE or FALSE and TRUE or TRUE✓✓
TRUE or FALSE or TRUE

The answer is TRUE.✓ (5)

5.2 5.2.1 A memory location, which holds a value that cannot be changed.
(Or any other valid definition)

5.2.2 A memory location, which holds a value that can be changed.
(Or any other valid definition)

5.2.3 Dividing a computer program into smaller programs.
(Or any other valid definition)

5.2.4 A program that responds to user actions or other events triggered from within the program.
(Or any other valid definition)

5.2.5 A record is composed of related fields or columns and appears as rows within a data table.
(Or any other valid definition)
(5 × 1) (5)
[10]

QUESTION 6

6.1 6.1.1
```
SELECT *✓
FROM tblHusband✓
WHERE Province = “Limpopo”✓
```
(3)

6.1.2
```
SELECT✓ FullName, English, Maths, Biology, History, PCP,
English + Maths + Biology + History + PCP✓ AS TOTAL,✓
Province✓
FROM tblHusband✓
```
(Award mark for the correct use of the SUM function) (5)

6.2 6.2.1 To calculate the sum of two or more numbers

6.2.2 To find the lowest number in a series

6.2.3 To find the highest number in a series

(3 × 1) (3)
[11]

QUESTION 7

7.1 Database✓
ANY ONE of the following features:
- Databases store data or information in tables✓
- Each table contains a lot of fields and records
- Each table contains a primary key
- Can contain many tables
- Usually supplied as a package
- Can be linked by using joins/relationships

Array✓
ANY ONE of the following features:
- Temporary storage✓
- Stores a number of values under a single name
- Can only store data of one data type

Text-file✓
ANY ONE of the following features:
- Made up of a long string of alphanumeric characters✓
- A sequential file
- Unstructured

(Mark the name of a data structure and only one of the features) (Any 3 × 2) (6)

7.2 7.2.1 Selection sort will find the smallest number✓ in the whole list and put it in the first position. In this instance, it will start with 2,✓ then look for the second smallest number✓ in the remaining numbers, which, in this instance, will be 21.✓ Selection sort will do so until all the numbers are sorted.
(Or any other valid explanation) (4)

7.2.2 Binary search;✓ Selection Search;✓ Linear Search;✓ Jump Search;✓ Interpolation Search;✓ Exponential Search;✓ Sublist Search;✓ Fibonacci Search;✓ Ubiquitous Binary Search✓
(Any 2 × 1) (2)
[12]

QUESTION 8

8.1
- Resizing objects on a screen
- Stopping the program before it finishes
- Entering invalid data
- Missing data on an input source (Any 3 × 1) (3)

8.2 To ensure the integrity✓ during data transmission or storage✓
(Or any other valid explanation) (2)

8.3 8.3.1 Logical error
8.3.2 Syntax error
8.3.3 Semantic error
(3 × 2) (6)
[11]

QUESTION 9

9.1

INPUT	PROCESS	OUTPUT
Age✓	If Age >=18✓ then strMessage = 'You are allowed to vote'✓ Else strMessage = 'You are NOT allowed to vote'✓ End If	strMessage✓

(5)

9.2 9.2.1 Nothing

9.2.2 When the start value is greater than the end value, the step value must be a negative number.

9.2.3 Add()

(3 × 2) (6)

[11]

QUESTION 10

10.1 10.1.1 A thoughtless user is one who responds without “thinking” about what is in front of him/her on the computer.

(Or any other valid explanation) (2)

10.1.2
- Use exception handling
- Use Input Data Masks
- Check ALL input data before processing
- Check that selected action is valid before processing
- Use programming “assertions”
- Reduce free input areas by using restricted choices, i.e. checkboxes, radio buttons and select dropdown lists.

(Or any other valid explanation) (Any 5 × 1) (5)

10.2
- Range check
- Type check
- Length check
- Format check
- Presence check

(Any 3 × 1) (3)

[10]

TOTAL SECTION B: 75
GRAND TOTAL: 100

Copyright reserved